

IC-31 INTRODUCTION TO AC/DC ELECTRONICS LAB

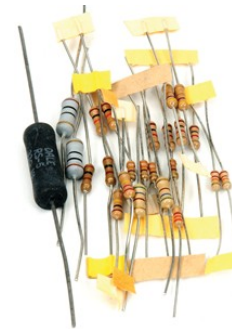
Rev 1-8-2023 (originally by AQ)

31.1 OBJECTIVE

Learn how to use the Circuit Board and Measuring Instruments.

31.2 EQUIPMENT

1. EM-8656 AD/DC Electronics Lab
2. Student Multimeter
3. Voltage Sensor and Current sensor
4. 2 Batteries
5. Wires and Resistors



31.3 STRUCTURE OF THE AC/DC CIRCUIT BOARD:

The structure of the Circuit Board (EM-8656 AD/DC Electronics Lab) as shown in the Figure 1. We define the names of each spring connector in the circuit board as shown in the figure. In the rectangular areas of Figure 1, the springs are connected in pairs, oriented perpendicular to each other and named C and S. In a given rectangular area, the two springs should be treated equally in circuit connection. For example, connecting to C1 means connecting to S1 also. Therefore the names C1 and S1 are interchangeable in the following circuit build procedures. Similarly for C2 and S2, and the others.

The two springs in each of the square boxes (e.g. S1 and C1 etc.) are connected together, and so S1 and C1 are essentially the same point, but S1 and S2 (and the rest) are not connected together – we will make these connections in our circuit. The way the connections are made is: take the wire and push it in-between the coils of the springs in

such a way that the wire protrudes on both sides, see Fig. 2. This ensures that the wire is touching the spring at four points, which makes a good connection.

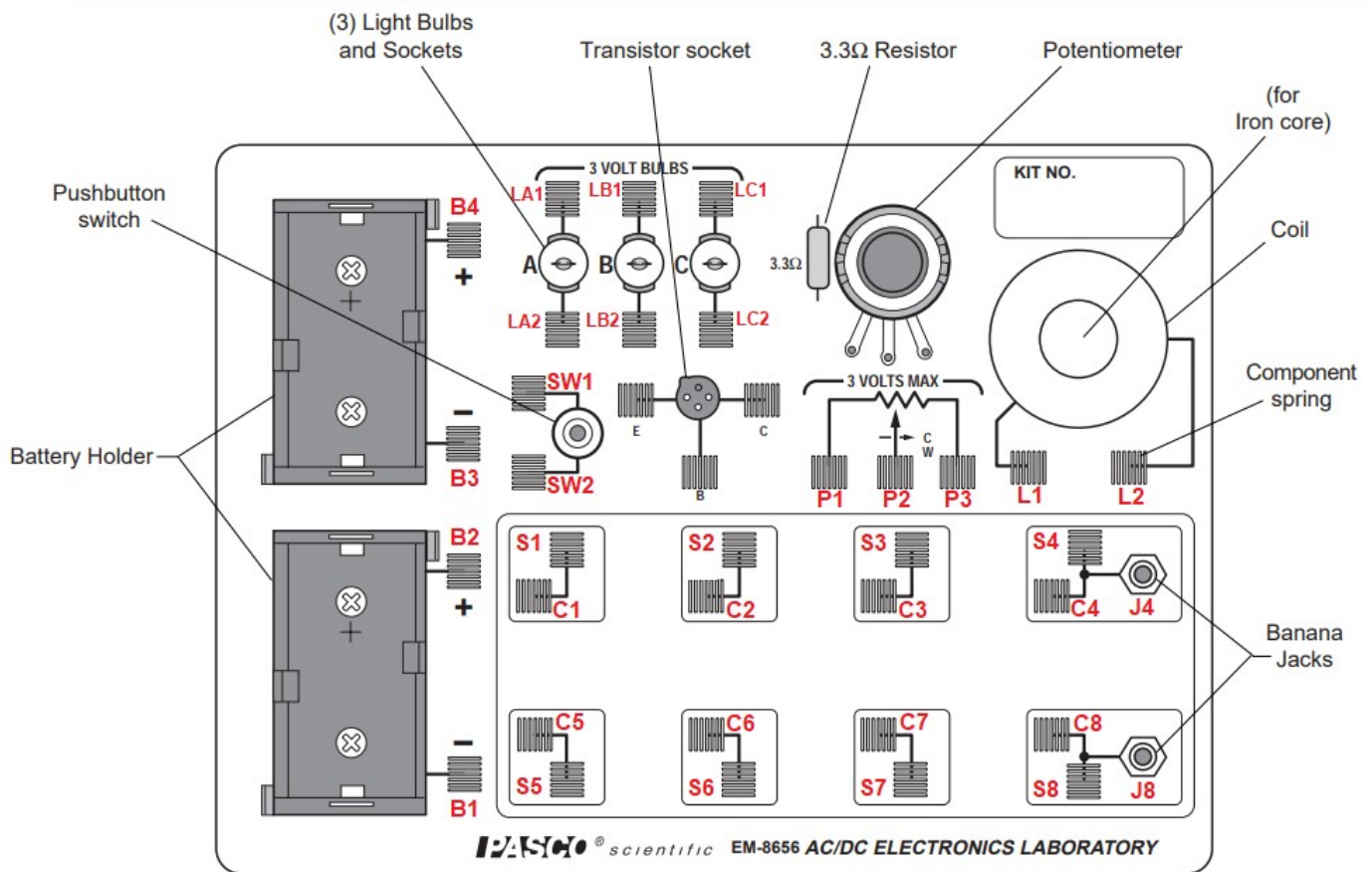


Figure 1: Definition of the names of each spring connector

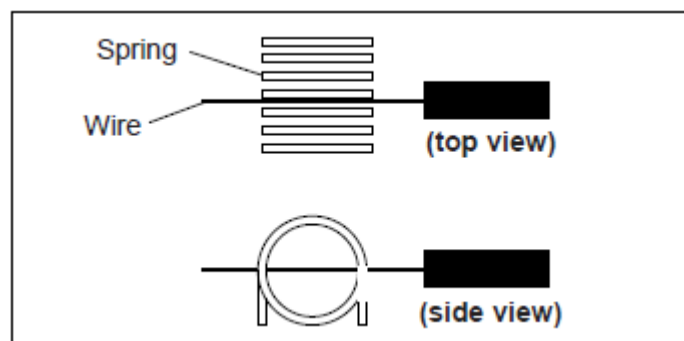


Figure 2: How to attach wires to springs

31.4 LEARN HOW TO USE MULTIMETER:

A Multimeter, as the name suggests, is an instrument that can measure many different parameters by adjusting a switch. The Multimeters that we have in the Labs and in the Lab Kits can measure:

1. Continuity
2. Resistance
3. DC Voltage ($\text{—}| \text{—}| \text{—}|$ symbol above the voltage range)
4. AC Voltage ($\sim$ symbol above the voltage range)
5. DC Current ($\text{—}| \text{—}| \text{—}|$ symbol above the current range)
6. Temperature
7. Battery Test
8. Diode Test



For all measurements, insert the BLACK test lead banana plug into the **COM** jack, and the RED test lead banana plug into the **V•Ω•mA** jack. Only for measuring high current, upto 10 A, the RED wire goes in the **10A** jack (we will not be needing this in our Labs).

Turn the rotary switch to the parameter and its appropriate range, then touch the tips of the test probes to the locations where the parameter is to be measured. If you are not sure of the expected range of values, start with selecting the highest range.

The Voltage is measured ACROSS a circuit element. That is why the Multimeter has a very high resistance when used to measure the voltage (so that minimum current flows through it, and it does not alter the circuit).

The current is measured THROUGH the circuit. In this case, the circuit is broken, and the multimeter is placed so that it re-connects the broken part. The current now flows through the multimeter. The Multimeter in this case has a very low resistance, so that it does not alter the circuit.

The resistance should be measured when the resistor is not part of the circuit. This can be done by taking the resistance out of the circuit, or by taking out at least one leg of the resistor.

The HOLD button freezes / unfreezes the display.

1. Continuity Test:

- 1.1. A multimeter has the Continuity Test mode (|||||).
- 1.2. Continuity is the presence of a complete path for current flow. A circuit is complete when its switch is closed.

- 1.3. A digital multimeter's Continuity Test mode can be used to test switches, fuses, electrical connections, conductors and other components. A good fuse, for example, should have continuity.
- 1.4. A multimeter emits an audible response (a beep) when it detects a complete path.
- 1.5. The beep, an audible indicator, permits technicians to focus on testing procedures without looking at the multimeter display.
- 1.6. When testing for continuity, a multimeter beep's based on the resistance of the component being tested. That resistance is determined by the range setting of the multimeter. Examples:
- 1.7. If the range is set to 400.0 Ω , a multimeter typically beeps if the component has a resistance of 40 Ω or less.
- 1.8. If the range is set 4.00 k Ω , a multimeter typically beeps if the component has a resistance of 200 Ω or less.
- 1.9. The lowest range setting should be used when testing circuit components that should have low-resistance value such as electrical connections or switch contacts.

2. Resistance Measurement:

- 2.1 **Coding on the Resistors:** Resistors are color coded to indicate their value. The color coding can be seen in Figure 3, where each color represents a number. To read the color coding, note the colors of the four bands. The first and second numbers (i.e. colors) are the first and second digits of the resistance value. The third number (color) is the multiplier in power of ten. The fourth color is the tolerance (or error) in the value of the resistance. As an example, a resistor colors are: Orange-Yellow-Red-Gold. The color coding shows Orange and Yellow to be 3 and 4, which makes it 34. The Red is 2, which means 10^2 . The resistance is then 34×10^2 ohms. Gold indicates that this value is within $\pm 5\%$ of its actual resistance. Similarly: Blue-Red-Green-Silver becomes 62×10^5 ohms $\pm 10\%$.

<u>First Three Bands</u>				<u>Fourth Band</u>	
Black	0	1st digit2nd digitNumber of zeroesTolerance	None	$\pm 20\%$	
Brown	1		Silver	$\pm 10\%$	
Red	2		Gold	$\pm 5\%$	
Orange	3		Red	$\pm 2\%$	
Yellow	4				
Green	5				
Blue	6				
Violet	7				
Gray	8				
White	9				
ex: Orange - Green - Red - Gold					
	352			$5\% = 3500 \pm 5\% \Omega$	

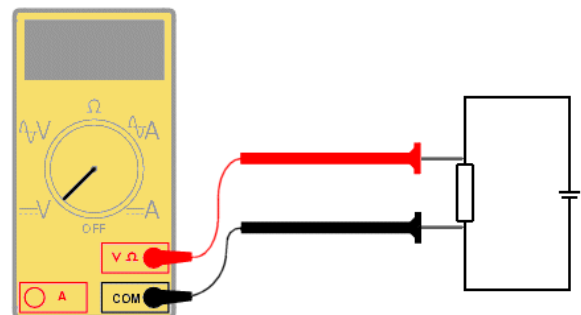
Figure 3: Resistance Color Coding Chart

- 2.2 **Measure resistance when components are not connected in a circuit:** It is always advisable **not** to measure the resistance of an item that is in a circuit. It is always best to make the measurement of the resistance on its own out of the circuit. If a measurement is made in-circuit, then all the other components around it will have an effect. Any other paths that will allow current to pass will affect the readings, making them inaccurate to some degree.
- 2.3 **Remember to ensure the circuit under test is not powered on:** Under some circumstances it is necessary to measure resistance values actually on a circuit. When doing this it is **very important to ensure the circuit is not powered on**. In out circuits this means that the battery is either not connected in the AC/DC Box, or, if the battery is directly connected to the switch then the switch is OPEN, i.e. in the OFF state. Not only will any current flowing in the circuit invalidate any readings, but should the voltage be high enough, the current resulting could damage the multimeter.
- 2.4 **Leakage path through fingers can alter readings in some cases.** When making some resistance measurements it is necessary to hold a resistor or component onto the multimeter test probes. If high resistance measurements are being made the leakage path through the fingers can become noticeable.
- 2.5 **Select the required range:** The digital multimeter needs on and the required range selected. The range selected should be such that the best reading can be obtained. Normally the multimeter function switch will be labelled with the maximum resistance reading. Choose the one where the estimated value of resistance will be under but close to the maximum of the range. In this way the most accurate resistance measurement can be made.
- 2.6 **Make the measurement:** With the multimeter ready to make the measurement the probes can be applied to the item that needs to be measured. The range can be adjusted if necessary.
- 2.7 **Turn off the multimeter:** Once the resistance measurement has been made, the multimeter can be turned off to preserve the batteries. It is also wise to turn the function switch to a high voltage range. In this way if the multimeter is used to again for another type of reading then no damage will be caused if it is inadvertently used without selecting the correct range and function.

3. Voltage Measurement:

Voltage measurements look at the potential difference between two points. Procedures of voltage measurement with a multimeter:

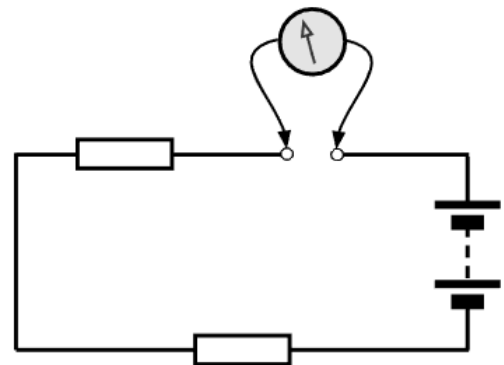
- 3.1 Before current measurement, check the whether the multimeter's socket is correct.



- 3.2 At the beginning the measurement, use maximum range to protect the meter. If needed reduce the range gradually.
- 3.3 Switch the multimeter to the voltage ranges. It is best to select a range higher than the expected voltage so that there is no chance of the meter being overloaded and damaged.
- 3.4 Check that the test leads are plugged into the correct sockets. Many multimeters have different sockets for different types of measurement so it is worth checking the correct ones have been chosen before making the measurement. Usually a meter will be provided with two leads, one black, and the other red. The black one is normally taken as the negative one. It is connected to the negative or "common" socket on the meter. The red one is connected to the positive socket.
- 3.5 When making the measurement, the positive lead should be connected to terminal which is expected to have the higher electric potential. If the leads are connected the wrong way around a negative voltage will be displayed.
- 3.6 With the multimeter connected, power can be applied to the circuit. The multimeter switches can then be changed to reduce the value of the range. This is done until the greatest number of significant digits can be read on the meter. In this way the most accurate reading is obtained.

4. Current Measurement:

- 4.1 Current measurement requires that the circuit in which the current is flowing has to be broken and the multimeter inserted into the circuit.
- 4.2 In order that the multimeter does not alter the operation of the circuit when it is used to measure current, the resistance of the meter must be as low as possible. Therefore resistances of meters used to measure current are normally very low.



- 4.3 Before current measurement, check the circuit loop is not short (circuit has at least one resistor). And multimeter socket is correct. At the beginning the measurement, use maximum range or socket first. If needed reduce the range or socket. When the measurement finished, return to the meter's original setting.
- 4.4 Insert the probes into the correct connections. Often one labelled common into which the black probe is normally placed. The other probe should be entered into the correct socket for the current measurement to be made.

- 4.5 Set main selector switch on the meter switch to the correct measurement type, (i.e. DC current) and range for the measurement to be made.
- 4.6 When the measuring the current, optimize the range for the best reading. If possible enable all the leading digits to not read zero, and in this way the greatest number of significant digits can be read.
- 4.7 Once the reading is complete, it is a wise precaution to place the probes into the voltage measurement sockets and turn the range to maximum voltage. This will prevent the meter from damage if the meter is accidentally connected without thought.

31.5 LEARN HOW TO USE VOLTAGE AND CURRENT SENSORS:

The Voltage and Current Sensors (and also the Voltage and Current Probes) can measure the voltage and current when connected to Capstone. Depending on their type, they may either be connected to the 850 Interface with wires, or wirelessly to Capstone through bluetooth. In the Lab Kits we have the Wireless Voltage and Current Sensors. The maximum range for the Wireless voltage sensors is 15 volts, and for the Current Sensor it is one ampere.

1. If the Voltage and Current Sensors are wired, connect them both to Science Workshop 850, and connect the 850 to the computer.
2. If the Voltage and Current Sensors are of the wireless type, connect them through Bluetooth to the computer (you do not need the Science Workshop 850).
3. Turn on Capstone Software. Turn on the Voltage Sensor and Current Sensor. Click Hardware Setup. Capstone will start detecting the Sensors. Select them when Capstone detects them. Make sure the numbers match. Click Hardware Setup again to close it.
4. In the graphs, select "Sensor Data". Select Voltage or Current (as the case may be) on the Y-axis, and time on the X-axis.
5. With the sensors not connected to the circuit, press 'Tare' to 'zero' the sensors. This will set the voltage and current values in the sensors to be set to zero.
6. Connect the prongs of the Voltage or Current Sensor to the relevant springs on the AC/DC Box.
7. Press 'Record' in Capstone. Press the Switch. You should now see graph of current or voltage in Capstone. Stop Recording after a few seconds. Select a range of data and obtain the mean value.

31.6 PROCEDURE

A. Resistance Measurements (Figure 6)

1. Select 4 resistors of different values. Label them as R1, R2, R3 and R4. Do not select any resistor of less than 100 ohms. Note their colors and coded values in Table 1.
2. Attach one resistor at a time between the springs S3 and S7. Measure their resistance and note in Table 1.
3. Calculate and note the percent difference in the measured and coded values.

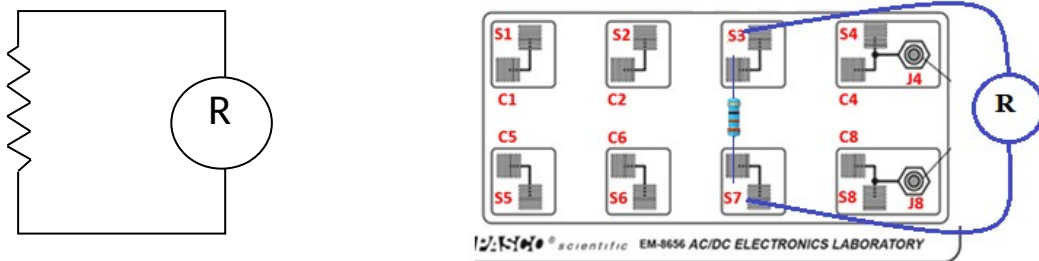


Figure 6: Resistance Measurement, Circuit Diagram and Setup on the AC/DC Box

B. Battery Voltage (Figure 7)

4. Attach one D-Cell in the top Battery Holder (i.e. between B3 and B4), keeping the positive side towards B4.
5. Set the Multimeter to DC Voltage, with a range around 5 volts. Connect one prong of the Multimeter to B4 and the other to B3. Note the voltage of the battery.

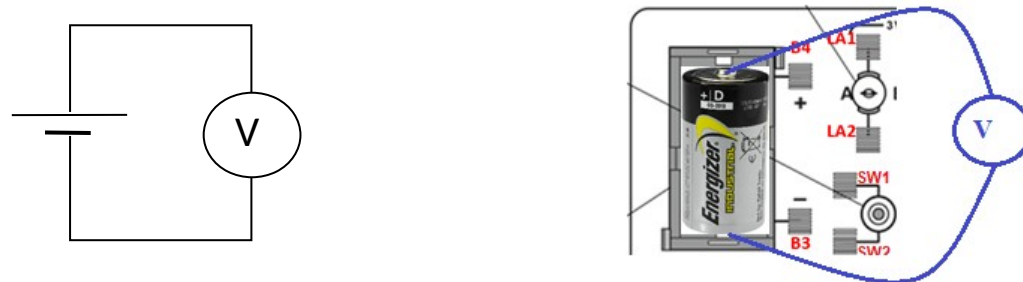


Figure 7: Battery Voltage, Circuit Diagram and Setup on the AC/DC Box

C. Voltage Across Resistors (Figure 8)

6. Take some wires and connect: B4 to SW1 (one end of the Push-Button Switch). Connect SW2 (the other end of the Push Button Switch) to S3. Connect S7 to S6. Connect S6 to B3. Connect one resistor between S3/C3 and S7/C7.
7. When you press the Push Button Switch, the circuit will be complete, and current will flow from positive of the battery to B4 – to the Switch – to S3 – through the resistor to S7 – to S6 – to B3 – to the negative of the battery.

8. Set the Multimeter to read DC voltage, range about 5 volts. Connect one of its prongs to S3 and other to S7. Now press the Switch and note the reading of the voltage across the resistance. Repeat with four other resistances. And note the values in table 2.

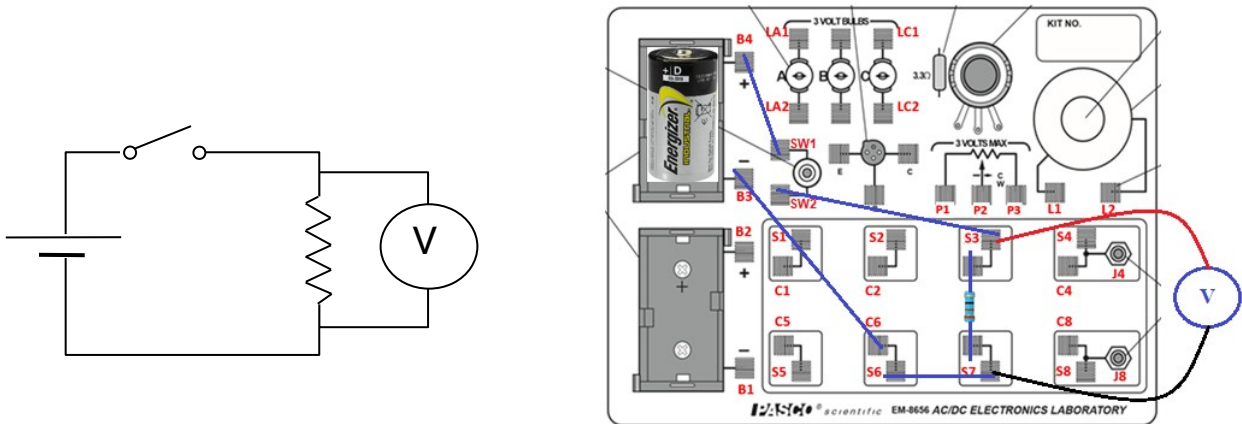


Figure 8: Voltage across Resistor, Circuit Diagram and Setup on the AC/DC Box

C. **Current Through Resistors (Figure 9)**

9. Now remove the Multimeter. Set it on DC Current. Select the highest value of current as the range. Remove the wire between S7 and S6, and connect the two prongs of the Multimeter in between these two (S6 and S7). Now the Multimeter is in series with the circuit, and ALL current through the resistor will also flow through it.
10. Press the Switch and note the value of the current. If you do not get a value, reduce the range on the ammeter and measure current. (Always start with the highest range. The ammeter can easily be damaged by a current that is beyond its selected range). Note the values in Tabele 2.

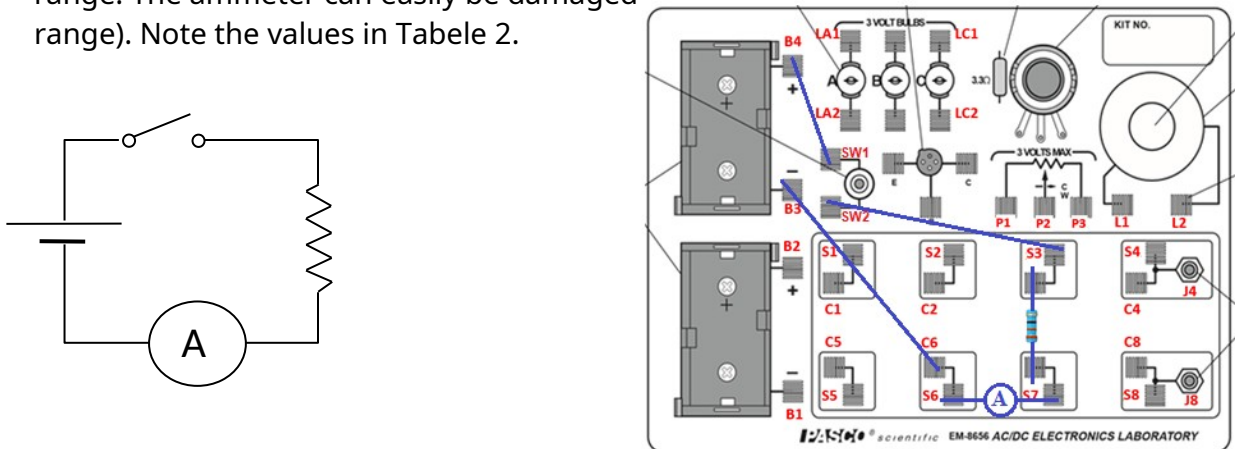


Figure 9: Current Through Resistor, Circuit Diagram and Setup on the AC/DC Box

E. **Using Sensors for Voltage and Current (Figure 10)**

1. Take out the Multimeter and connect the Current Sensor in its place.

2. Connect the Voltage and Current Sensors. Turn on Capstone Software. Turn on the Voltage Sensor and Current Sensor. Click Hardware Setup. Select the Sensors when Capstone detects them. In the graphs, select "Sensor Data". Select Voltage or Current on the Y-axis, and time on the X-axis.
3. With the sensors not connected to the circuit, press 'Tare' to 'zero' the sensors.
4. Connect the prongs of the Voltage Sensor to S3 and S7. Connect the prongs of the Current Sensor to S7 and S6.
5. Press 'Record' in Capstone. Press the Switch. You should now see graph of current or voltage in Capstone. Stop Recording. Select a range of data and obtain the mean value.
6. Repeat measurements with the same four resistors as before.
7. Enter the data in Table 2.

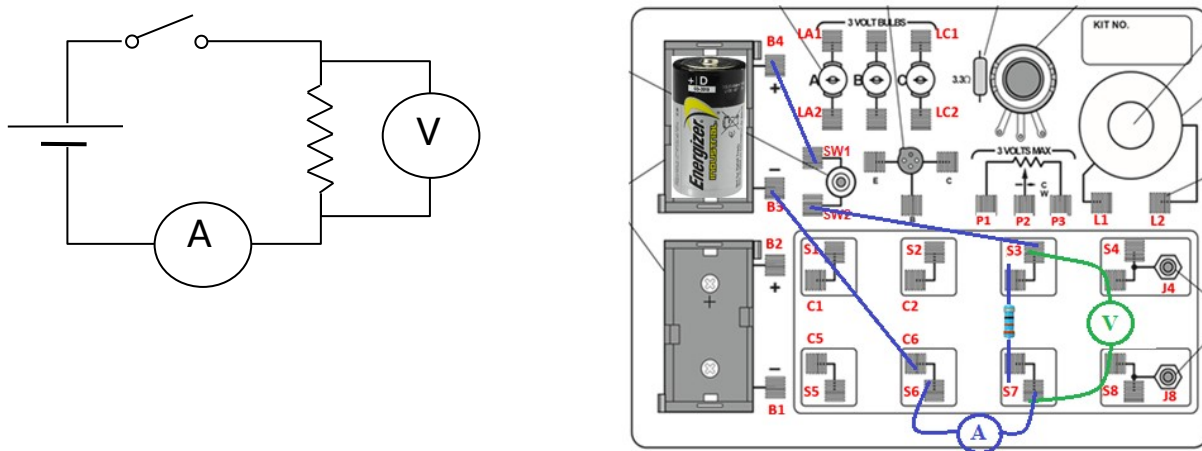


Figure 10: Voltage and Current Sensors, Circuit Diagram and Setup on the AC/DC Box

31.7 LIGHTS IN CIRCUITS

In this part of the Lab, we will study the effect of having lights in series and parallel connections with the battery, and the function of the potentiometer.

1. Remove all previous connections. Make the circuit as shown in Fig. 11, and note the brightness of the lamp.
2. Reconnect the circuit as shown in Fig. 12. The two lamps are now in series. Note their brightness.
3. Reconnect the circuit as shown in Fig. 13. The two lamps are now in parallel. Note their brightness.
4. Reconnect the circuit as shown in Fig. 14. Rotate the knob of the potentiometer and see the effect on the brightness of the lamp.

Taking data is not required in this part. Write what you observed in the Results section. There may be questions about what you would have observed here in the Post Lab Quiz.

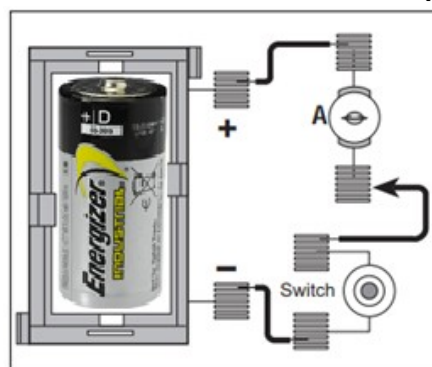
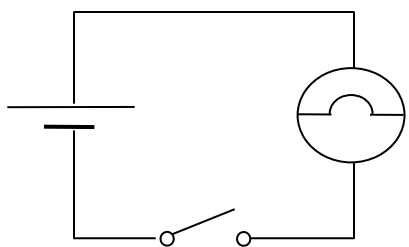


Fig. 11: One Lamp, Circuit Diagram and Setup on the AC/DC Box

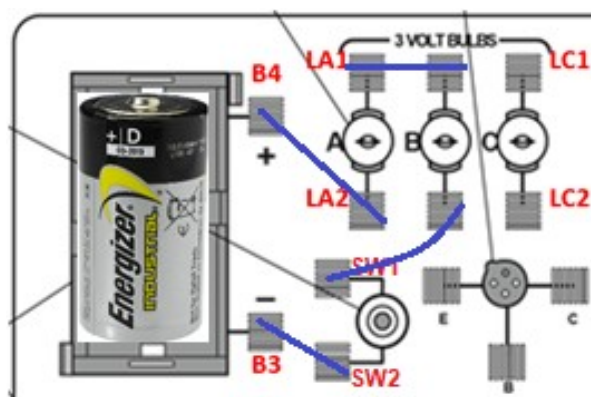
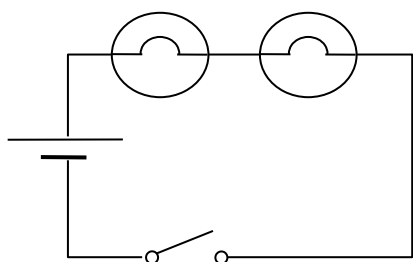


Fig. 12: Two Lamps in Series, Circuit Diagram and Setup on the AC/DC Box

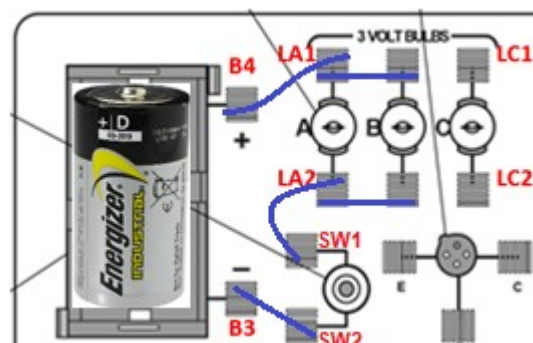
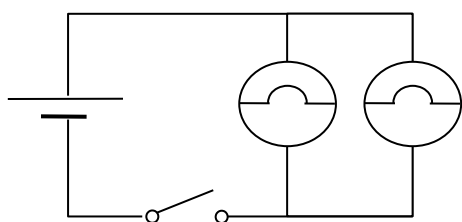


Fig. 13: Two Lamps in Parallel, Circuit Diagram and Setup on the AC/DC Box.

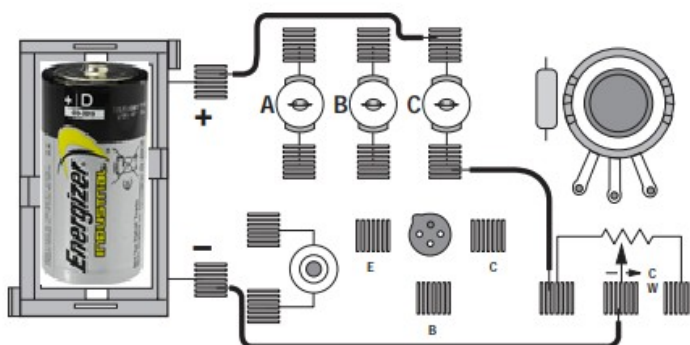
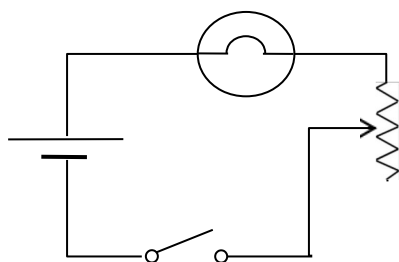


Fig. 14: Lamp through Potentiometer, Circuit Diagram and Setup on the AC/DC Box

31.8 PRECAUTIONS

1. Attach the black lead to the COM terminal in the Multimeter and the red lead to the other terminal. Always touch the circuit with the red lead closer to the battery positive side.
2. Make sure that the connection between the wire and the spring is good. The spring should pass through the spring and touch it at two points (see Fig. 2).
3. Always start with the multimeter setting at the higher end before testing the current or voltage, then reduce it to get better results. This is especially important for current measurements, as the multimeter can easily burn out if the actual current is much higher than the scale being measured.
4. Sometimes it is hard to distinguish between colors on the resistors. In this case, use the Multimeter to measure the resistance to make sure you have it right.
5. The resistances can be attached with any of the two ends towards the positive of the battery.
6. Do not select very high resistance, else the current will be too small to measure.
7. Attach the battery in the correct direction.
8. Remember that S1 and C1 are two springs, but they are the same point in the circuit. Where it says S1, it also means C1. Same for S2 to S8.

31.9 REPORT FORM

DATA

Table 1: Resistances

Resistance No	Color codes	Resistor Value from Color Codes	Measured Value of Resistance	Percent Error in Resistance
R1				
R2				
R3				
R4				

Table 2: Voltages and Currents.

Battery Voltage from Procedure B: _____

Resistance No	By Using Multimeter		By Using Wireless Sensors	
	Measured Voltage Across Resistor	Measured Current Through Resistor	Measured Voltage Across Resistor	Measured Current Through Resistor
	Units	Units	Units	Units
R1				
R2				
R3				
R4				

Note; The Multimeter and Sensors should give you same (or very close) values. If they are more than about 10% different, check your connections.

Lights In Circuits (Write your observations here)

31.10 REPORT:

For In-Class lab:

Submit the following in the Lab Report in Canvas		
1.	Filled up Table 1	10
2.	Filled up Table 2	10
3.	Screenshot of capstone showing voltage and current being measured by the Multimeter / Sensors	5x2 = 10
4.	Discussion / Conclusion / What you learnt from this Lab	10
	total	40

For Take-Home Lab-Kit lab:

Submit the following in the Lab Report		
Show During Lab Time		Points
1.	Use your camera to show your setup for Resistance Measurement.	5
2.	Use your camera to show your setup for Current or Voltage Measurement using Multimeter or Sensor.	5
3.	Share your screen to show current or voltage measurement being made by Capstone.	5
Report (Upload in Canvas)		
4.	Filled up Table 1	10
5.	Filled up Table 2	10
6.	At least three pictures, showing the following being measured: Resistance, Voltage, Current (by either multimeter or sensors)	3*5 = 15
7.	Screen-shot of capstone showing voltage or current being measured.	5
8.	Discussion / Conclusion / What you learnt from this Lab	10
	Total	65

31.11 ADDITIONAL INFORMATION:

<https://www.youtube.com/watch?v=bF3OyQ3HwfU>

31.12 POINTS TO THINK ABOUT

1. What would happen if you touch the red lead to the negative terminal of the battery and the black lead to the positive terminal?
2. Why is the multimeter resistance very small when used as an ammeter, and very high when used as a voltmeter.
3. What would be the error if the multimeter resistance is not very low or very high in the two cases of the point number 2?
4. You would note that the lamps are brighter when they are in parallel (as compared to when they are in series). Why is that so?
5. If you had two small lamps, and one glowed brighter than the other (for the same circuit configuration), which one would have the bigger resistance in its filament?